

■ Midterm Exam

Introduction to C Programming Essentials

Borg Al Arab Technological University • Spring 2026

Data Types, Variables & I/O	15
Functions	5
Operators	10
Conditions (if / switch)	10
Loops + Break & Continue	10
Total	50

■ Instructions

- Choose the correct answer (a, b, c, or d).
- Questions are deep, tricky, and include common mistakes, precedence, and undefined behavior.
- Answers are at the very end — do NOT look until you finish all 50 questions.

Section 1: Data Types, Variables, Input & Output (15 Questions)

Q1. Which declaration is completely incorrect?

- a) `float x = 3.14f;`
- b) `double y = 3.14;`
- c) `int 1num = 10;`
- d) `char name[50] = "Omar";`

Q2. The size of `double` in C is:

- a) 4 bytes
- b) 8 bytes
- c) 2 bytes
- d) 1 byte

Q3. What happens if you write `scanf("%d", num);` without `&`?

- a) Compile error
- b) Runtime error or undefined behavior
- c) Works normally
- d) Prints the value

Q4. `printf("%.2f", 15.567);` will print:

- a) 15.57
- b) 15.56
- c) 15.5
- d) 15.567

Q5. Which line is required to use both `printf` and `scanf`?

- a) `#include`
- b) `#include`
- c) `#include`
- d) `#include`

Q6. `char ch = 'A'; printf("%d", ch);` will print:

- a) A
- b) 65
- c) 97
- d) Error

Q7. The best way to read a string that may contain spaces is:

- a) `scanf("%s", str);`
- b) `fgets(str, 100, stdin);`
- c) `scanf("%[^\n]", str);`
- d) Both b and c

Q8. `float f = 3.14159; printf("%.3f", f);` will print:

- a) 3.142
- b) 3.141
- c) 3.14
- d) 3.14159

Q9. The difference between `const int MAX = 100;` and `#define MAX 100` is:

- a) `const` is stored in memory, `#define` is not
- b) `#define` is replaced before compilation
- c) `const` cannot be changed, `#define` can
- d) Both a and b are correct

Q10. After `scanf("%d", &x);`, if the user presses Enter, what happens in the next `scanf("%c", &ch);`?

- a) Reads the next character
- b) Reads the leftover `\n` (very common bug)
- c) Program freezes
- d) Prints error

Q11. Which is a correct declaration for a char variable?

- a) `char c = 'AB';`
- b) `char c = 65;`
- c) `char c = "A";`
- d) `char c = A;`

Q12. `printf("%f", 3.14);` (without precision) will print:

- a) 3.140000
- b) 3.14
- c) 3
- d) Error

Q13. In most modern compilers, the size of `int` is:

- a) 2 bytes
- b) 4 bytes
- c) 8 bytes
- d) Depends on the system

Q14. `scanf("%f", &gpa);` is used to read:

- a) Integer only
- b) Float or double
- c) String
- d) Char

Q15. Which code causes undefined behavior?

- a) `int x; printf("%d", x);`
- b) `int x = 5; printf("%d", x);`
- c) `char c = 'A';`
- d) `#include`

Section 2: Functions (5 Questions)

Q16. A function that does not return any value has the type:

- a) `int`
- b) `void`
- c) `float`
- d) `char`

Q17. To call a function before defining it, you must:

- a) Write a function prototype
- b) Put the function inside main
- c) It is impossible
- d) Use `#include`

Q18. `int add(int a, int b) { return a + b; }` → `add(3, 4)` returns:

- a) 3
- b) 4
- c) 7
- d) Error

Q19. The return type of the `main()` function is always:

- a) `void`
- b) `int`
- c) `double`
- d) `char`

Q20. The main benefit of using functions is:

- a) Increasing errors
- b) Code reusability + cleaner code
- c) Reducing speed
- d) Making code unreadable

Section 3: Operators (10 Questions)

Q21. `10 % 3` equals:

- a) 3
- b) 1
- c) 0
- d) 10

Q22. `int x = 5; int y = ++x;` → value of y is:

- a) 5
- b) 6
- c) 7
- d) Error

Q23. `int x = 5; int y = x++;` → value of y is:

- a) 5
- b) 6
- c) 7
- d) Error

Q24. `5 == 5 && 3 > 1` returns:

- a) 0
- b) 1
- c) true
- d) false

Q25. `! (5 > 3)` returns:

- a) 1
- b) 0
- c) true
- d) false

Q26. Which operator has the highest precedence?

- a) `&&`
- b) `==`
- c) `++` (prefix)
- d) `+`

Q27. `x += 5;` is equivalent to:

- a) `x = x + 5;`
- b) `x = 5;`
- c) `x = x - 5;`
- d) `x = 5 + x;`

Q28. `5 / 2` when both are `int` equals:

- a) 2.5
- b) 2
- c) 3
- d) 0

Q29. `int x = 10; x = x + 5 * 2 % 3;` → final value of x:

- a) 15
- b) 13
- c) 12
- d) 11

Q30. `int x = 5; printf("%d", x++ + ++x);` (undefined behavior, but most compilers give):

- a) 11
- b) 12
- c) 10
- d) 13

Section 4: Conditions & If / Switch (10 Questions)

Q31. `switch` statement works with:

- a) float
- b) int and char only
- c) double
- d) string

Q32. In `if (x = 5)`, what actually happens?

- a) It compares x with 5
- b) Assignment (value becomes 5 → true)
- c) Syntax error
- d) false

Q33. `int day = 3; switch(day) { case 3: printf("Wed"); break; } prints:`

- a) Wed
- b) 3
- c) Invalid
- d) Nothing

Q34. `else if` can be replaced by:

- a) switch only
- b) nested if
- c) both
- d) none

Q35. `if (x > 5) if (x < 15) printf("A");` (no braces) will execute:

- a) Only "A"
- b) A and B
- c) Syntax error
- d) C

Q36. The `default` case in switch is:

- a) Mandatory
- b) Optional
- c) Must be the first case
- d) Always needs break

Q37. `if (x > 0 && x % 2 == 0)` checks for:

- a) x is positive and even
- b) x is negative
- c) x is odd
- d) x equals 0

Q38. Correct way to check if a number is zero:

- a) `if (x == 0)`
- b) `if (x = 0)`
- c) `if (!x)`
- d) Both a and c

Q39. In `switch (x) { case 1: ... break; default: ... }`, if `x = 2` it will:

- a) Execute case 1
- b) Execute default
- c) Execute nothing
- d) Error

Q40. `x > y ? x : y` is an example of:

- a) Ternary operator
- b) Normal if
- c) switch
- d) Error

Section 5: Loops + Break & Continue (10 Questions)

Q41. Which loop guarantees the code runs at least once?

- a) `for`
- b) `while`
- c) `do-while`
- d) `switch`

Q42. `for(int i=1; i<=5; i++)` runs:

- a) 4 times
- b) 5 times
- c) 6 times
- d) Never

Q43. `int i=5; do { printf("%d", i); i++; } while(i < 5);` prints:

- a) 5
- b) Nothing
- c) 5 6
- d) Infinite loop

Q44. `for(int i=1; i<=6; i++) { if(i==3) continue; printf("%d ", i); }` prints:

- a) 1 2 3 4 5 6
- b) 1 2 4 5 6
- c) 1 2 4 5
- d) 3 only

Q45. `for(int i=1; i<=10; i++) { if(i==6) break; printf("%d ", i); }` prints:

- a) 1 2 3 4 5 6
- b) 1 2 3 4 5
- c) 6
- d) Infinite

Q46. `int i=0; while(i < 10) { i++; if(i%2==0) continue; printf("%d ", i); }` prints:

- a) 1 3 5 7 9
- b) 2 4 6 8 10
- c) 1 2 3 4 5
- d) 1 3 5 7 9 10

Q47. In nested loops `for(i=1;i<=2;i++) for(j=1;j<=2;j++)`, the inner loop runs how many times total?

- a) 2
- b) 4
- c) 3
- d) 1

Q48. `break` inside a loop:

- a) Exits the entire loop
- b) Skips the current iteration
- c) Continues normally
- d) Restarts the loop

Q49. `continue` inside a `while` loop:

- a) Exits the loop
- b) Skips the rest of the code and goes back to the condition
- c) Stops the program
- d) Prints error

Q50. `int sum=0; for(int i=1; i<=10; i+=2) sum += i;` → final value of sum:

- a) 25
- b) 55
- c) 30
- d) 10

■ Answer Key

Check only after solving all 50 questions!

Section 1 – Data Types, Variables & I/O				
Q1. C	Q2. B	Q3. B	Q4. A	Q5. B
Q6. B	Q7. D	Q8. A	Q9. D	Q10. B
Q11. B	Q12. A	Q13. B	Q14. B	Q15. A
Section 2 – Functions				
Q16. B	Q17. A	Q18. C	Q19. B	Q20. B
Section 3 – Operators				
Q21. B	Q22. B	Q23. A	Q24. B	Q25. B
Q26. C	Q27. A	Q28. B	Q29. B	Q30. B
Section 4 – Conditions & If/Switch				
Q31. B	Q32. B	Q33. A	Q34. C	Q35. A
Q36. B	Q37. A	Q38. D	Q39. B	Q40. A
Section 5 – Loops + Break & Continue				
Q41. C	Q42. B	Q43. A	Q44. B	Q45. B
Q46. A	Q47. B	Q48. A	Q49. B	Q50. A

Good luck, Omar! You will ace the midterm! ■